ROOF DITCH MATERIAL CRACKING—SERVICE TIPS

Article No. 02-21-4

FORD: 1997-2003 F-150
1999-2003 WINDSTAR

LINCOLN: 2002 BLACKWOOD

ISSUE

Some vehicles may exhibit repeated cracking of the paint or sealer in the roof ditch due to incorrect repair procedures.

ACTION

Read this TSB in its entirety BEFORE beginning any repairs. Inspect and repair as outlined in the service procedures listed below. Do not use any other materials or procedures to repair this issue.

SERVICE PROCEDURE

NOTE

READ THIS REPAIR PROCEDURE COMPLETELY BEFORE PERFORMING ANY SERVICE ACTIONS. USE THE PROCEDURE FOR THE PAINT BRAND USED IN YOUR SHOP.

Application Tips

- Use 1 300 ml cartridge per seam. Do not change cartridges in the middle of the seam.
- When inserting the cartridge in the application gun keep the gun pointed upward.
- When attaching the mixing nozzle, make sure the attaching nut is tight. This will prevent air from being sucked in creating bubbles. Never cut off the mixing tip as it provides the correct back pressure for correct flow.
- When filling the channel, keep the flow as continues as possible.
- If you see bubbles in the seam after application, pop the bubbles with a pin before the material dries.
- Paint can be applied in 60 minutes, as there is no shrinkage.
- Clean uncured sealer with wax and grease remover followed by wiping with a dry cloth.
- Work time at 70° F is 8-9 minutes, with full cure in 24 hours.

Steps Common To All Repairs

To reduce the length of the article, all procedures except the first one will be shortened using an abbreviated text to indicate 7 common steps needed to complete each procedure. The first procedure is complete and unabbreviated.

AKZO - SIKKENS- NATIONAL RULE (Tech Support 1-800-618-1010)

1. Remove original sealant materials from the roof ditch area using and air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with M600 wax and grease remover.
4. Mix and apply 1 medium coat (1.2-1.4 mils) of Autocoat L V Epoxy primer to all bare metal sand-through areas in the roof ditch.
5. Allow the primer to cure 2 hours at 68° F (20° C).
6. Lightly scuff sand the primer with P400-grit sand paper or red Scotchbrite pad.
7. Blow off the roof ditch with compressed air; wipe the repair area with M600 wax and grease remover, and tack off to remove loose dust and debris.

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COMMON TASKS TO ALL REPAIRS

- Mask off all areas on the roof ditch that could be affected by spills.

NOTE
DO NOT USE MASKING TAPE ALONG THE INSIDE OF THE OF THE ROOF DITCH CHANNEL. ANY SEALANT THAT COVERS THE TAPE DURING THE APPLICATION IS NOT REMOVABLE WITHOUT DAMAGING THE CURED SEALANT.

- Position the vehicle so it is level front to rear.
- Stand the Motorcraft TA-15 Roof Ditch Sealer in a vertical position (nose up) for more than 30 minutes before use to allow air bubbles to rise to the top, where they can be purged.

NOTE
IF THE TRAPPED AIR BUBBLES ARE NOT REMOVED FROM THE CARTRIDGE, THERE WILL BE AIR BUBBLES IN THE SEAM, WHICH WILL CAUSE THE PAINT TO POP.

- Apply masking tape to form a dam where the windshield pinch weld meets the front end of the roof ditch. The tape dam will prevent the sealant from flowing into the pinch weld. Remove the tape after the sealer has cured.

NOTE
IF THE WINDSHIELD IS IN PLACE, PROP THE WINDSHIELD MOULDING UP SO IT IS LEVEL AND USE AS A DAM.

- Insert the Motorcraft TA-15 Roof Ditch Sealer in the application gun in an upward position.
- Remove the plastic caps and squeeze out a small amount until both parts dispense equally. Keeping the gun in an upward position will push any air out first
- Attach the mixing tip keeping the gun tipped upwards.
- Gun out a small amount of material in an upward position to force the air out of the tube, then tilt the nozzle down and run out another small amount to verify air purging.

NOTE
AIR BUBBLES MAY BE PRESENT AS THE MATERIAL PASSES THROUGH THE MIXING TIP. AFTER THERE ARE NO MORE BUBBLES YOU ARE READY TO BEGIN THE REPAIR.

- Starting from the back of the vehicle and moving forward, apply the sealant to the center of the channel allowing it to flow into the entire seam area. Keep the mixing tip in the puddle of sealer as you move down the channel.
- Reduce the volume of sealant being applied as the curvature of the roof panel is being approached. Stop dispensing the sealant as soon as it begins to flow down the slope and allow it to flow naturally.
- No tooling should be needed, but if necessary, shape within 3-5 minutes after application.
- Allow 45-60 minutes cure time before paint application. If sealer has cured longer than 2 hours, scuff the surface with red Scotchbrite pad.

NOTE
THIS NEXT STEP IS VERY IMPORTANT AS TO PREVENT FLEX CRACKING ON THE SURFACE.

8. Mix and apply 3 coats of Colorbuild (1st coat thin, followed by 2 medium coats, 1.2-1.5 mils per coat) of primer surfacer over the seam sealer. Add flex agent to Colorbuild then mix as normal. Allow the flexed primer to flash 5-10 minutes between coats. Bake 30-45 minutes at 140° F.
9. Sand with P220-P320 grit sand paper then final sand with P400-P500 grit sand paper. Blow off the roof ditch with compressed air; wipe the repair area with M600 wax and grease remover, and tack off to remove loose dust and debris.
10. Mix and apply Autobase Plus basecoat (add 10% Standard Hardener to basecoat before reducing) and spray to hiding. Allow 3-7 minutes flash between coats. Allow 20 minutes before final tack off and applying clearcoat.
11. Mix Autoclear III (add flex agent to clear, then mix hardener and activator as normal) and apply 2 medium coats, 1.2-1.5 mils per coat. Let flash 3-7 minutes between coats and bake for 30 minutes at 140° F panel temperature.
12. Demask the vehicle, and deliver to the customer.

AKZO-SIKKENS RULE 45 AND 1151 (Tech Support 1-800-618-1010)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with M600 wax and grease remover.

4. Mix and apply 1 medium coat (1.2-1.4 mils) of Autocoat L V Epoxy primer to all bare metal sand-through areas in the roof ditch.
5. Allow the primer to cure 2 hours at 68° F (20° C).
6. Lightly scuff sand the primer with P400-grit sand paper or red Scotchbrite pad.
7. Blow off the roof ditch with compressed air; wipe the repair area with M600 wax and grease remover, and tack off to remove loose dust and debris.

SEE COMMON TASKS TO ALL REPAIRS

NOTE
THIS NEXT STEP IS VERY IMPORTANT AS TO PREVENT FLEX CRACKING ON THE SURFACE.

8. Mix and apply 4 coats of Colorbuild 2.1/2.8 Primer (1st coat thin, followed by 2 medium coats, then 1 mist coat 1.0-1.3 mils per coat) of primer surfacer over the seam sealer. Add flex agent to Colorbuild 2.1/2.8 then mix as normal. Allow the flexed primer to flash 5-10 minutes between coats. Bake 30-45 minutes at 140° F.
9. Sand with P220-P320 grit sand paper then final sand with P400-P500 grit sand paper. Blow off the roof ditch with compressed air; wipe the repair area with M600 wax and grease remover, and tack off to remove loose dust and debris.
10. Mix and apply Autobase Plus basecoat (add 10% Autoclear II LV Hardener to basecoat before reducing) and spray to hiding. Allow 3-7 minutes flash between coats. Allow 20 minutes before final tack off and applying clearcoat.
11. Mix Autoclear II (add flex agent to clear, then mix hardener and activator as normal) and apply 2 medium coats, 1.2-1.5 mils per coat. Let flash 3-7 minutes between coats and bake for 30 minutes at 140° F panel temperature.
12. Demask the vehicle, and deliver to the customer.

BASF R-M (Tech Support 1-800-825-3000)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with PreKleano wax and grease remover.
4. Mix and apply 1 to 1.5 wet coats (.8-1.0 mil) of R-M DE15 epoxy primer to all bare metal sand-through areas in the roof ditch.
5. Allow the primer to cure 2 hours at 68° F (20° C).
6. Lightly scuff sand the primer with 220-grit sand paper or red Scotchbrite pad.
7. Blow off the roof ditch with compressed air; wipe the repair area with PreKleano wax and grease remover, and tack off to remove loose dust and debris.

SEE COMMON TASKS TO ALL REPAIRS

NOTE
THIS NEXT STEP IS VERY IMPORTANT AS TO PREVENT FLEX CRACKING ON THE SURFACE.

8. Mix and apply 2-3 medium wet coats (1.5-2.0 mils) of R-M DP20/DF25 Flex agent as a primer surfacer over the seam sealer. Allow the flexed primer to flash 5-10 minutes between coats. Bake 30-45 minutes at 140° F.
9. Sand the cured DP20 with 500 or finer grit sand paper. Blow off the roof ditch with compressed air; wipe the repair area with PreKleano wax and grease remover, and tack off to remove loose dust and debris.
10. Mix and apply R-M Diamont basecoat per label instructions and spray to hiding. Allow 10 minutes flash between coats. Allow 20 minutes before applying clearcoat.
11. Mix R-M DC92 and apply 2 wet coats, 2.0 mils. Let flash 5-10 minutes between coats and bake for 30 minutes at 140° F panel temperature.
12. Demask the vehicle and deliver to the customer.

BASF Glasurit (Tech Support 1-800-825-3000)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with 541-5 wax and grease remover.

4. Mix and apply 1 to 1.5 wet coats (.8-1.0 mil) of Glasurit 801-1552 epoxy primer to all bare metal sand-through areas in the roof ditch.
5. Allow the primer to cure 2 hours at 68° F (20° C).
6. Lightly scuff sand the primer with 220-grit sand paper or red Scotchbrite pad.
7. Blow off the roof ditch with compressed air; wipe the repair area with 541-5 wax and grease remover, and tack off to remove loose dust and debris.

SEE COMMON TASKS TO ALL REPAIRS

NOTE
THIS NEXT STEP IS VERY IMPORTANT AS TO PREVENT FLEX CRACKING ON THE SURFACE.

8. Mix and apply 2-3 medium wet coats (1.5-2.0 mils) of Glasurit 285-50 with 521-111-flex agent as a primer surfacer over the seam sealer. Allow the flexed primer to flash 5-10 minutes between coats. Bake 30-45 minutes at 140° F.
9. Sand the cured primer with 500 or finer grit sand paper. Blow off the roof ditch with compressed air; wipe the repair area with 541-5 wax and grease remover, and tack off to remove loose dust and debris.
10. Mix and apply Glasurit 55 line basecoat per label instructions and spray to hiding. Allow 10 minutes flash between coats. Allow 20 minutes before applying clearcoat.
11. Mix Glasurit 923-455 clearcoat and apply 2 wet coats, 2.0 mils. Let flash 5-10 minutes between coats and bake for 30 minutes at 140° F panel temperature.
12. Demask the vehicle, and deliver to the customer.

PPG Deltron System (Tech Support 1-440-572-6111)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 120-180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with DX330 wax and grease remover.
4. Mix and apply 1 medium coat (1.0-1.5 mils dry) of DP90LF Epoxy primer with DP401LF catalyst mixed at 2:1 by volume and to all bare metal sand-through areas in the roof ditch.

NOTE
ALLOW 30-MINUTE INDUCTION PERIOD TO OBTAIN PERFORMANCE PROPERTIES BEFORE USING DPLF.

5. Allow the primer to flash 1 hour at 68° F (20° C)

SEE COMMON TASKS TO ALL REPAIRS

6. Mix and apply 1 coat of SX1056 flexible 2K sealer with DCX8 high solids hardener and DT reducer mixed at 4:1:2 (1.0-1.5 mils dry) of primer surfacer over the seam sealer. Allow the primer to flash 5-10 minutes between coats. Sand with P400-P500 grit sand paper. Blow off the roof ditch with compressed air; wipe the repair area with DX330 wax and grease remover, and tack off to remove loose dust and debris. Do not bake at this point.
7. Mix DBC Deltron Basecoat with DT reducer and DX57 basecoat activator mixed 1:1 + 5% and apply 2-3 coats or till hiding is achieved. Allow 3-7 minutes flash between coats. Allow 15 minutes before final tack off and applying clearcoat.
8. Mix DCU2042 Low VOC Speed Clear with DCX61 hardener and DT reducer mixed 4:1:1 and apply 2 medium coats, 2.0-2.5 mils. Let flash 0-5 minutes between coats and bake for 20 minutes at 140° F panel temperature.
9. Demask the vehicle and deliver to the customer.

PPG Global System (Tech Support 1-440-572-6111)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 120-180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with D837 wax and grease remover.

4. Mix D8099 Anti-Corrosion Etch primer with D8299 primer catalyst mixed at 1:1 by volume and apply 1 medium coat (.3-.4 mils) of and to all bare metal sand-through areas in the roof ditch.
5. Allow the primer to flash 30 minutes to 1 hour at 68° F (20° C).

SEE COMMON TASKS TO ALL REPAIRS

6. Mix and apply 1 coat of SX1056 flexible 2K sealer with D897 catalyst and appropriate reducer mixed at 4:1:2 (1.0-1.5 mils dry) of primer surfacer over the seam sealer. Allow the primer to flash 30 minutes.
7. Sand if necessary with P400-P500 grit sand paper. Blow off the roof ditch with compressed air; wipe the repair area with D837 wax and grease remover, and tack off to remove loose dust and debris.
8. Mix BC Global basecoat with reducer and D888 basecoat activator at 1:1: + 5% and apply 2-3 coats and spray to hiding. Allow 3-7 minutes flash between coats. Allow 15 minutes before final tack off and applying clearcoat.
9. Mix D893 Low VOC performance clear with D884 high solid hardener and the appropriate reducer mixed at 3:1:1 and apply 2 coats, 2.0-2.5 mils. Let flash 0-5 minutes between coats and bake for 20 minutes at 140° F panel temperature.
10. Demask the vehicle and deliver to the customer.

Spies Hecker® Low VOC Recommendation (Tech Support 888-371-3313, www.performancecoatings.dupont.com)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with Permahyd® Silicone Remover 7095 wax and grease remover.
4. Apply 2 medium wet coats (Recommended film build: 0.4 mils in 2 coats) of Priomat® 1:1 Primer 4130 Transparent to all bare metal sand-through areas in the roof ditch. Mix ratio: 4130/1 Part, 3689/1 Part. Allow material to air dry 4 hours at 68° F (20° C) Do not scuff or sand.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and apply 2-3 coats of Permacron® 2.1 Surfacer 5155 + 30% Permasolid® Elastic Additive 9050 (Spies Hecker® TDS 341.4) (Recommended film build max: 3.0 mils in 2 coats) Mix ratio: 5155/1 Part, 9050/30% Then mix the above products 4:1+20-30% with: 3195/1 part, 3395/20-30%. Air dry 3 hours or flash 5 -10 minutes, then force dry 30 minutes at 140° F. Sand with red 3M Scotch-Brite® Pad.
6. Blow off the roof ditch with compressed air; wipe the repair area with Permahyd® Silicone Remover 7095 wax and grease remover, and tack off to remove loose dust and debris.
7. Apply Permacron® 293 / 295 Series Basecoat + Hardener to hiding. (0.6-1.0 mil) (Spies Hecker® TDS 970.10 & 970.13) Mix Ratio: Permacron® Basecoat/1 Part, 3193/15%, 3054/40% spray to hiding. Let material air dry 25 minutes. Do not scuff or sand.
8. Apply 2 medium wet coats of Permacron® 2.1 Clear Coat 8095 + Permasolid® Elastic Additive 9050 (Recommended film build: 2.0 - 2.4 mils in 2 coats) Mix Ratio (Spies Hecker® TDS 111.8) 8095/1 part, 9050/30%. Then mix the above products 2:1+5-10% with: 3193/1 Part, 3395/5-10%. Air dry 4-6 hours or bake 30-40 minutes at 140° F.
9. De-mask the vehicle and deliver to the customer.

Spies Hecker® National Rule Recommendation (Tech Support 888-371-3313, www.performancecoatings.dupont.com)

1. Remove original sealant materials from the roof ditch area using and air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with Permacron® Silicone Remover 7010 wax and grease remover.
4. Apply 2 medium wet coats of Priomat® 1:1 Primer 4130 Transparent to all bare metal sand-through areas in the roof ditch. (Recommended film build: 0.4 mils in 2 coats) Mix ratio: 4130/1 Part, 3689/1 Part. Allow material to air dry 4 hours at 68° F (20° C). Do not scuff or sand.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and Apply 2-3 coats of Permacron® Vario Surfacer 8590 + 30% Permasolid® Elastic Additive 9050 (Spies Hecker® TDS 988.3) (Recommended film build: 2.5 - 3.5 mils in 3 coats) Mix ratio: 8590/1 Part, 9050/30%. Then mix the above products 3:1+20% with: 3315/1 part, 3363/20%. Allow material to air dry 3 hours or flash 5 -10 minutes then, force dry 30 minutes at 140° F. Sand with red 3M Scotch-Brite® Pad.
6. Blow off the roof ditch with compressed air; wipe the repair area with Permacron® Silicone Remover 7010 wax and grease remover, and tack off to remove loose dust and debris.
7. Apply Permacron® 293 / 295 Series Basecoat + Hardener to hiding. (0.6-1.0 mil) Mix Ratio (Spies Hecker® TDS 970.10 & 970.13) Permacron® Basecoat/1 Part, 3315/15%, 3054/40%.Let material air dry 25 minutes. Do not scuff or sand.
8. Apply 2 medium wet coats of Permacron® MS Clear Coat 8110 + Permasolid® Elastic Additive 9050 (Recommended film build: 2.0 - 2.4 mils in 2 coats) Mix Ratio (Spies Hecker® TDS 111.9) 8010/1 part, 9050/30%. Then mix the above products 3:1+5-7% with: 3315/1 Part, 3363/5-7%. Air dry overnight or bake 30 minutes at 140° F.
9. De-mask the vehicle and deliver to the customer.

Standox® Low VOC Recommendation (Tech Support 800-551-9296, www.performancecoatings.dupont.com)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with Standohyd® Low VOC Silicone Remover wax and grease remover.
4. Mix and apply 2 medium wet coats of Standox® Etching Reaction Primer to all bare metal sand-through areas in the roof ditch. (Recommended film build: 0.32 mill in 2 coats) Mix ratio (Standox® TDS 314) 13908/1 Part, 11166/1 Part. Allow material to air dry 4 hours at 68° F (20° C) Do not scuff or sand.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and Apply 2-3 coats of Standox® 2.1 Filler + 30% Standox® 2K Plasticiser (Recommended film build: Max. 3.0 mil in Max. 2 coats) Mix ratio (Standox® TDS 545) Volume (mixing stick A) 11999/1 Part, 15260/30%. Above is then mixed 2:1+30% with: 10089/15277- 1 Part, 10097/15269- Approximately 30% (to viscosity). Air dry 3 hours or flash 5 -10 minutes then force dry 30 minutes at 140° F. Scuff sand with red 3M Scotch-Brite® Pad.
6. Blow off the roof ditch with compressed air; wipe the repair area with Standohyd® Low VOC Silicone Remover wax and grease remover, and tack off to remove loose dust and debris.
7. Apply Standox® MSB Basecoat to hiding. (0.6 - 1.0 mil) (Standox® TDS 610, Important Remarks Section) Mix Ratio: Standox® color, Volume (mixing stick A) 1 Part, 10089/15277-20%. 11050/11395- 30-40% (to viscosity) Allow material to air dry 20-30 minutes. Do not scuff or sand.
8. Apply 2 medium wet coats of Standox® 2.1 Clear + 15-30% Standox® 2K Plasticiser (Recommended film build: 2.0 - 2.4 mils in 2 coats (Standox® TDS 577) Mix Ratio: 10020/1 Part, 15260/15-30%. Above is then mixed 2:1+5-10% with: 10089/15277-1 Part, 10097/15269- 5-10%. Allow the material to air dry 4 - 6 hours or bake 30-40 minutes at 140° F.
9. De-mask the vehicle and deliver to the customer.

Standox® National Rule Recommendation (Tech Support 800-551-9296, www.performancecoatings.dupont.com)

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with Standox® Silicone Remover Antistatic 11654 wax and grease remover.
4. Mix and apply 2 medium wet coats of Standox® Etching Reaction Primer to all bare metal sand-through areas in the roof ditch. (Recommended film build: 0.32 mill in 2 coats) Mix ratio (Standox® TDS 314) 13908/1 Part, 11166/1 Part. Allow material to air dry 4 hours at 68° F (20° C) Do not scuff or sand.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and apply 2-3 coats of Standox® 2K Nonstop Primer Filler + 30% Standox® 2K Plasticiser (Recommended film build: 2.4 - 3.6 mils in 2 - 3 coats) (Standox® TDS 535) Mix ratio: 13320 Volume (mixing stick A) 1 Part, 15260/30%, Above is then mixed 3:1+20%, 10089/15277-1 Part, 11182/20%. Air dry overnight or flash 10-15 minutes, then bake 30 minutes at 140° F. Scuff sand with red 3M Scotch-Brite® Pad.
6. Blow off the roof ditch with compressed air; wipe the repair area with Standox® Silicone Remover Antistatic 11654 wax and grease remover, and tack off to remove loose dust and debris.
7. Apply Standox® MSB Basecoat to hiding. (0.6 - 1.0 mil) (Standox® TDS 610, Important Remarks Section) Mix Ratio: Standox® color, Volume (mixing stick A) 1 Part, 10089/15277-20%. 11050/11395- 30-40% (to viscosity) Allow material to air dry 20-30 minutes. Do not scuff or sand.
8. Mix and apply 2 medium wet coats of Standocryl® 2K MS Clear + 15-30% Standox® 2K Plasticiser (Recommended film build: 2.0 - 2.4 mils in 2 coats) Mix Ratio (Standox® TDS 563) 11743/1 Part 15260/15-30%, above is then mixed 3:1+10-20%. 10089/15277-1 Part, 10097/15269-10-20% (to viscosity) Allow material to air dry overnight or flash 5-10 minutes, then bake 30 minutes at 140° F.
9. De-mask the vehicle and deliver to the customer.

Dupont California Recommendation 800-3DuPont (800-338-7668), www.performancecoatings.dupont.com

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with 3900S First Klean® wax and grease remover.
4. Mix and apply 1 medium wet coat of DuPont® Vinyl Wash Primer to all bare metal sand-through areas in the roof ditch. (Recommended film build: 0.4 - 0.6 mils in 1 coat) Mix ratio: Product Volume Weight (cumulative pt) 1305S 1 Part 208.1 grams 1306S 1 Part 410.5 grams Allow material to air dry between 1 - 2 hours. Do not scuff or sand.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and apply 3 medium wet coats of DuPont® 2K High Build Primer (Recommended film build: 4 - 6 mils in 2 coats) Mix ratio: Product Volume Weight (cumulative qt) 3240S 2 Parts 1014 grams 3205S 1 Part 1337 grams Allow material to air-dry: 1 - 2 hours at high humidity (50% RH or higher). 3 - 4 hours at low humidity (30% RH or lower). Force or force dry: 30 minutes at 140° F. Sand with red 3M Scotch-Brite® Pad.
6. Blow off the roof ditch with compressed air; wipe the repair area with 3901S Final Klean® wax and grease remover, and tack off to remove loose dust and debris.
7. Mix and apply ChromaBase® Basecoat to hiding. (0.5 - 2.0 mils) Mix Ratio: Product Volume ChromaBase color 1 Part 7160S, 7175S, 7185S, 7195S 1 Part 12305S 1 Oz. per ready-to-spray qt. Allow material to air dry 20 minutes. Do not scuff or sand.
8. Apply 2 medium wet coats of ChromaClear® Multi-Use 2100S (Recommended film build: 1.8 - 2.2 mils in 2 coats) Mix ratio: Product Volume Weight (cumulative qt) 2100S 3 parts 630.5 grams 2105S 1 part 841.2 grams. Allow material to air dry 16 hours or bake 30 minutes at 140° F followed by 4 hour cool down.

NOTE

DUE TO THE DIFFERENCE IN REDUCER WEIGHTS, USE THE FOLLOWING REDUCER WEIGHTS TO FINISH THE CUMULATIVE SPRAYABLE QUARTS. 2165S 1006.9 GRAMS 2175S 1040.5 GRAMS 12375S 1096.2 GRAMS.

9. Demask the vehicle and deliver to the customer.

Dupont National Rule Recommendation 800-3DuPont (800-338-7668), www.performancecoatings.dupont.com

1. Remove original sealant materials from the roof ditch area using an air gun with a chisel bit.

2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.

3. Wipe the repair area with 3900S First Klean® wax and grease remover. Mix and apply 1 medium wet coat of DuPont® Vinyl Wash Primer to all bare metal sand-through areas in the roof ditch. (Recommended film build: 0.4 - 0.6 mils in 1 coat) Mix ratio: Product Volume Weight (cumulative pt) 1305S 1 Part 208.1 grams 1306S 1 Part 410.5 grams Allow material to air dry between 1 - 2 hours. Do not scuff or sand.

SEE COMMON TASKS TO ALL REPAIRS

4. Mix and apply 3 medium wet coats of DuPont® 2K Urethane Primer- Filler + 2350S (Recommended film build: 3.2 - 4.8 mils in 3 coats) Mix ratio: Product Volume Weight (cumulative qt) 32420S 5 Parts 798 grams 12305S 1 Part 908 grams 12375S 1 Part 1000 grams 2530S 1 Part 1104 grams Allow material to air dry overnight or flash 5 -10 minutes then force dry 30 minutes at 140° F. Sand with red 3M Scotch-Brite® pad.

5. Blow off the roof ditch with compressed air; wipe the repair area with 3901S Final Klean® wax and grease remover, and tack off to remove loose dust and debris.

6. Mix and apply ChromaBase® Basecoat to hiding. (0.5 - 2.0 mils) Mix Ratio: Product Volume ChromaBase color 1 Part 7160S, 7175S, 7185S, 7195S 1 Part 12305S 1 Oz. per ready-to- spray qt. Allow material to air dry 20 minutes. Do not scuff or sand.

7. Mix and apply 2 medium wet coats of ChromaPremier® Productive Clear (Recommended film build: 1.8 - 2.2 mils in 2 coats) Mix Ratio: Product Volume Weight (cumulative qt) 72200S 9 Parts 476.8 grams 12305S 3 Parts 651.2 grams 2350S 1 Parts 706.8 grams 12375S 25-30 % 902.6 grams Allow material to air dry 16 hours or bake 20 - 30 minutes at 140° F followed by 2 hour cool down.

8. Demask the vehicle and deliver to the customer.

Sherwin-Williams National Rule Recommendation (Tech Support 1-800-526-6704 / Fax Back Services 1-800-824-0631)

1. Remove defective Seam Sealant from entire length of the roof ditch rail. Strip all paint and primer from this repair area. This can easily be accomplished using an angle grinder and a 3M Roloc Clean 'N Strip Disc (part number 07466). Solvent clean this area with SC 158 surface cleaner.

2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.

3. Wipe the repair area with SC 158 surface cleaner.

4. Mix and apply 1 medium coat of NP75 (.8 to 1.3 mils.) a true DTM primer surfacer. This is a non / iso product that offers great physicals to bare steel along with fast cure rate. This product can be used in open shop environments. Allow this to dry 60 minutes, and abrade the primer surface with a red scotch brite pad.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and apply 2 coats of P30 Color Prime at up to 3.5 to 4.0 mils. P30 is a premium urethane primer surfacer with exceptional flexibility, adhesion, and speed of cure. This can be mixed in a color that closely mimics the vehicle color. This can be easily sanded at 45 to 50 minutes with 400 grit sandpaper and finish sanded with 600 grit and prepped for top coat application. An optional step is that the roof panel repairs could be sealed using either S65 in 2K form, or S-P30 color seal.

6. Blow off the roof ditch with compressed air; wipe the repair area with SC 158 surface cleaner, and tack off to remove loose dust and debris.

7. Mix and apply Ultra Base B7 in 2K configuration (mixed 16-16-1 with UH904) is then applied to coverage over the repaired areas. This mix configuration offers excellent adhesion and flexibility. Let flash for 15 minutes.

8. Mix and apply CC930, Ultra Speed / Plus Performance Clearcoat applied in a WOW (limited flash) scenario. This cleacoat offers an exceptional appearance, flexibility, and dry times. Bake at 140° F for 15 minutes.

9. Demask the vehicle and deliver to the customer.

Sherwin-Williams Low VOC Recommendation (Tech Support 1-800-526-6704 / Fax Back Services 1-800-824-0631)

1. Remove defective Seam Sealant from entire length of the roof ditch rail. Strip all paint and primer from this repair area. This can easily be accomplished using an angle grinder and a 3M Roloc Clean 'N Strip Disc (part number 07466). Solvent clean this area with SC 158 surface cleaner.
2. Prep and clean the roof ditch area with 180 grit sand paper and blow off with compressed air.
3. Wipe the repair area with SC 158 surface cleaner.
4. Mix and apply 1 medium coat of NP75 (.8 to 1.3 mils.) a true DTM primer surfacer. This is a non / iso product that offers great physicals to bare steel along with fast cure rate. This product can be used in open shop environments. Allow this to dry 60 minutes, and abrade the primer surface with a red scotch brite pad.

SEE COMMON TASKS TO ALL REPAIRS

5. Mix and apply 2 coats of NP75. This can be easily sanded at 45 to 50 minutes with 400 grit sandpaper and finish sanded with 600 grit and prepped for top coat application.
6. Blow off the roof ditch with compressed air; wipe the repair area with SC 158 surface cleaner, and tack off to remove loose dust and debris.
7. Mix and apply Ultra Base B7 in 2K configuration (mixed 16-16-1 with UH904) is then applied to coverage over the repaired areas. This mix configuration offers excellent adhesion and flexibility. Let flash for 15 minutes.
8. Mix and apply CC920, Clearcoat applied in a WOW (limited flash) scenario. This cleacoat offers an exceptional appearance, flexibility, and dry times. Bake at 140° F for 15 minutes.
9. Demask the vehicle and deliver to the customer.

PART NUMBER	PART NAME
TA-15	Motorcraft Roof Ditch Sealer
TA-10	Motorcraft Application Gun

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 110000

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.